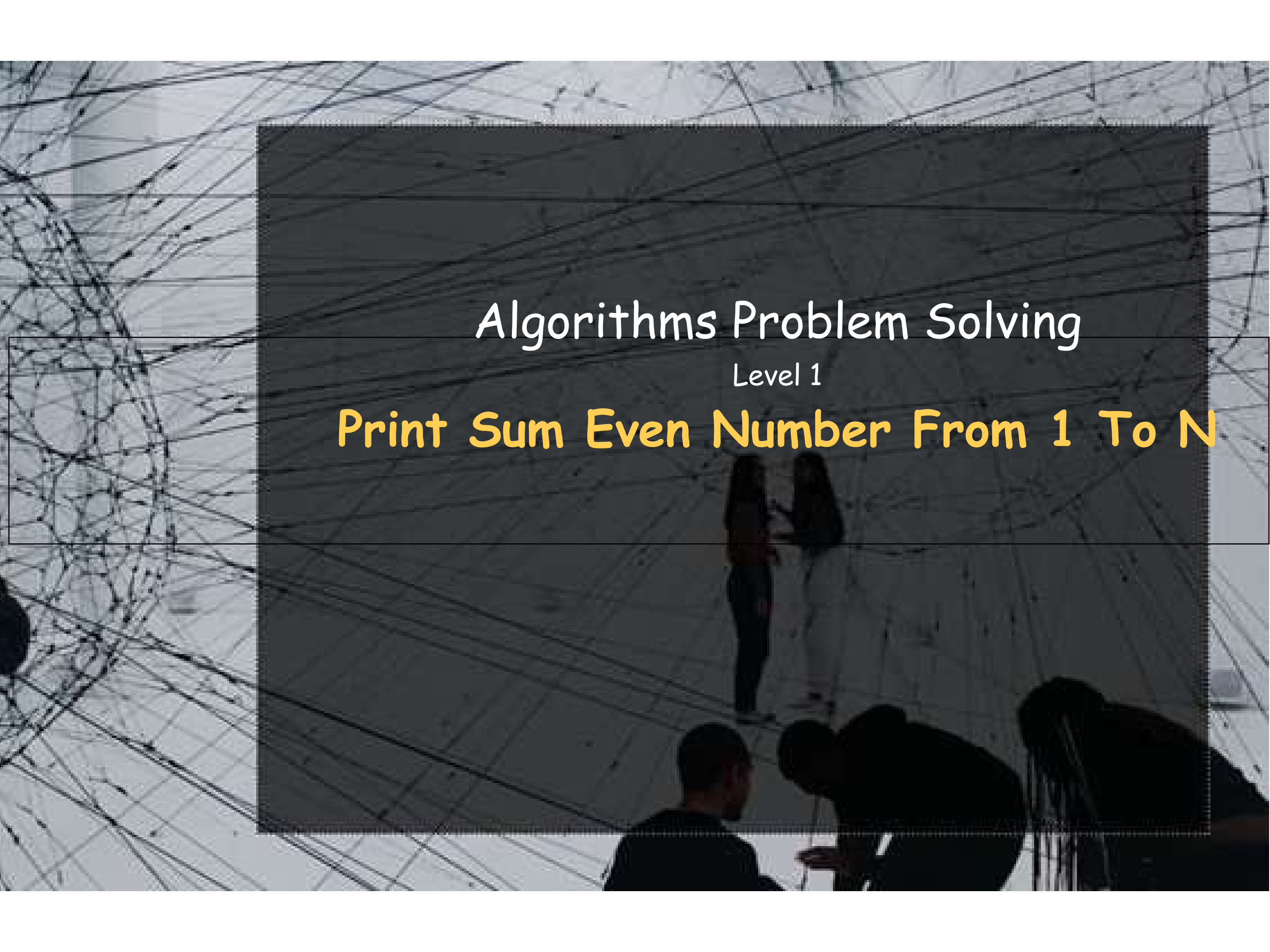


Algorithms Problem Solving

Level 1

Print Sum Even Number
From 1 To N



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Print Sum Even Number From 1 To N

Definition

- ◆ This program reads a number N and calculates the sum of all even numbers from 1 to N.
- ◆ It uses an `enum` to identify whether each number is odd or even.
- ◆ Only the numbers classified as `Even` are added to the total sum.

Explanation

- ◆ ``enOddOrEven`` is an enumeration with two values:
 - ``Odd = 1``
 - ``Even = 2``
- ◆ ``readNumb()`` reads N from the user and returns it.
- ◆ ``checkOddOrEven()`` determines whether a number is odd or even using ``% 2``.
- ◆ ``num % 2 != 0`` means the number is odd.
- ◆ Otherwise, the number is even.
- ◆ ``sumEvenNumbersFrom1ToN()`` starts ``sum`` at 0 and loops from 1 to N.
- ◆ For each number, it calls ``checkOddOrEven()``.
- ◆ If the returned value is ``Even``, the number is added to ``sum``.
- ◆ Finally, the function returns the total.
- ◆ ``sumEvenNumbersFrom1ToN(readNumb())`` reads N and passes the returned value directly to the summation function.

Steps in main()

1. Read the number N.
2. Set `sum` to 0.
3. Start the counter at 1.
4. Check whether the current number is even.
5. If it is even, add it to `sum`.
6. Increase the counter by 1.
7. Repeat until the counter reaches N.
8. Return the final sum.
9. Print the result.

💡 Example

◆ If the user enters:

$N = 10$

◆ The even numbers from 1 to 10 are:

2, 4, 6, 8, 10

◆ The program calculates:

$2 + 4 + 6 + 8 + 10 = 30$

➡ Result

Sum even numbers = 30

Note

◆ A number is even when dividing it by 2 leaves a remainder of 0.

◆ For example:

$`6 \% 2 = 0` \rightarrow \text{Even}$

$`7 \% 2 = 1` \rightarrow \text{Odd}$

⊘ Warning

- ◆ If N is less than 1, the loop does not execute and the result remains 0.



Comparison

▨ Even Number

- ✓ ``num % 2 == 0``
- ✓ Has no remainder when divided by 2.
- ✓ Examples: 2, 4, 6, 8, 10.

▤ Odd Number

- ✓ ``num % 2 != 0``
- ✓ Has a remainder when divided by 2.
- ✓ Examples: 1, 3, 5, 7, 9.

✦✦ Summary

- ✓ The program calculates the sum of even numbers from 1 to N.
- ✓ ``% 2`` is used to identify even and odd numbers.
- ✓ The ``enum`` represents the possible results of the check.
- ✓ Only numbers classified as ``Even`` are added to ``sum``.
- ✓ The same checking function can be reused for both odd and even calculations.

Thank you 😊